

National University of Computer and Emerging Sciences



Project Proposal
Farm Utility & Climate Control in Sheds (FUCCS)

for
Introduction to IoT

Course Instructor:	Prof. Ammar Haider
Subject:	Introduction to Internet of Things
Team Member(s):	Fayyez Farrukh BSE 2022 FAST NU LHR - 2217987 Haider Abbas Moazzam BSE 2022 FAST NU LHR - 2217862 Muhammad Arahim Bin Qaiser Nazir Butt BSE 2022 FAST NU LHR - 2216194
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Department of Computer Science
FAST-NU, Lahore, Pakistan

Introduction:

This project involves developing a complete IoT system designed specifically for modern poultry control sheds. Maintaining precise environmental conditions is critical in poultry farming, as sudden temperature spikes, high humidity, or incoming storms can lead to high bird mortality rates. By integrating real-time atmospheric sensing with cloud-based AI, this system moves beyond simple thermostats to predictive climate control. It monitors internal coop conditions and external barometric pressure to autonomously decide the most energy-efficient way to regulate the shed: using natural ventilation (actuated windows/vents), exhaust fans, or active air conditioning.

Goals:

1. **Weather Forecast:** Utilize barometric pressure trends to foresee incoming storms and automatically close vents to protect the flock from rain and drafts.
2. **Automated Coop Ventilation:** Use a mechanical servo to physically open or close shed vent windows based on internal heat and external weather forecasts.
3. **Energy Optimization:** Use a cloud-hosted AI model to predict temperature trends and decide the most cost-effective cooling method for the farm.
4. **Remote Farm Monitoring:** Provide a real-time web dashboard for farm managers to monitor shed climate statistics and manually override actuators via an MQTT broker.

Applications:

The primary application of the Farm Utility & Climate Control in Sheds (FUCCS) system is in commercial broiler poultry control sheds. Automating precise climate regulation is critical in this industry to maximize yield and minimize bird stress and mortality. Currently, climate control is a sensitive, manual responsibility. By automating this vital process, the FUCCS project is projected to significantly reduce monthly operational farm costs by approximately PKR 30,000.

Hardware Cost:

Item	QTY	Price (PKR)	Price (Gemini - usd)
ESP32 DevKit V1	1	1000/-	7
DHT22 (AM2302)	1	1372/-	3
BMP280 Sensor	1	150/-	4.5
TowerPro SG90 Servo Motor	1	300/-	3.5
5V 1-Channel Relay	1	800/-	2.5
Small DC Bulb with Holder	1	1600/-	6.5

5mm LED (Blue) + 220 Ω Resistor	1	16/-	0.5
Breadboard & Jumper Wires	1	430/-	5
Total		~5668/- pkr	~32.5/- usd (9000/- pkr)